Program: **Diploma in Food Processing Technology**	
Course Code: **6422A**	Course Title**: Cereals and Legumes**
Semester: **6**	Credits**: 4**
Course Category: **Open elective**	
Periods per week: **4(L: 4T:0 P: 0)**	Periods per semester: **60**

Course Objectives:

1. To understand the processing of rice and its byproduct utilization
2. To study about wheat, barley, corn, oats and millets and its processing method
3. To impart knowledge on composition and processing of legumes

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of cereals and legumes	Cereals and legumes

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the status of grain processing industries in India, rice varieties, nutrient composition, parboiling process with its physico-chemical changes, and the utilization of by-products like rice bran oil.	13	Understanding
CO2	Explain the nutrient composition and classification of wheat and barley, the milling process of wheat with its equipment, and the commercial production and nutritional importance of malt.	17	Understanding
CO3	Explain the processes of corn milling (dry and wet), oat composition, nutritional importance, processing, and the processing of major and minor millets, including pearl and finger millets	17	Understanding

CO4	Explain the nutrient composition and dietary importance of legumes, the impact of antinutritional factors in pulses on consumption, and the processing methods of soybean.	13	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the status of grain processing industries in India, rice varieties, nutrient composition, parboiling process with its physico-chemical changes, and the utilization of by-products like rice bran oil.**		
M1.01	Summarize the rice processing industry of India	5	Understanding
M1.02	Discuss the process of parboiling	5	Understanding
M1.03	Illustrate the byproduct utilization of rice processing	3	Understanding
Contents: Status of grain processing industries in India , rice- varieties, nutrient composition. Parboiling – physico chemical changes during parboiling – soaking – steaming – drying By product utilization – processing of rice bran oil			
CO2	**Explain the nutrient composition and classification of wheat and barley, the milling process of wheat with its equipment, and the commercial production and nutritional importance of malt.**		
M2.01	Classify wheat and explain its nutrient composition	6	Understanding
M2.02	Discuss the process of milling of wheat	5	Understanding

M2.03	Explain the nutrient composition and processing of Barley	5	Understanding
	Series Test –I	1	
Contents: Wheat - nutrient composition – classification of wheat, Milling of wheat - equipments used in the milling – break roll – reduction roll- purifier – plan sifter – scalping - scratch system. Barley – Nutrient composition – Commercial production of malt, Nutritional importance			
CO3	**Explain the processes of corn milling (dry and wet), oat composition, nutritional importance, processing, and the processing of major and minor millets, including pearl and finger millets**		
M3.01	Explain the process of milling of corn	5	Understanding
M3.02	Discuss the processing of oats, national importance, Composition	6	Understanding
M3.03	Summarize millets processing and types	6	Understanding
Contents: Milling of corn-dry milling and wet milling, Oats- composition, nutritional importance, processing, Millets- major and minor millets, Processing of millets-pearl and finger millet			
CO4	**Explain the nutrient composition and dietary importance of legumes, the impact of antinutritional factors in pulses on consumption, and the processing methods of soy**		
M4.01	Discuss about the nutrient composition and dietary importance of legumes	4	Understanding
M4.02	Discuss the antinutritional factors present in the pulses and its effect on consumption	4	Understanding
M4.03	Explain the processing of soya bean	4	Understanding
	Series Test–II	1	
Contents: Legumes- Nutrient composition, dietary importance Antinutritional factors present in the pulses and its effect on consumption Processing of soya bean			

Text / Reference

T/R	Book Title/Author
R1	Matz, Samuel A. “The Chemistry and Technology of Cereals as Food and Feed”. 2nd Edition, CBS, 2014

R2	Chakraverty, A. Post Harvest Technology of Cereals, Pulses and Oil Seeds , Third Edition, Oxford & IBH, 2008
R3	N.L.Kent, "Technology of Cereals", Elsevier, 1994.
R4	Delcour, Jan A. and R. Carl Hoseney. " Principles of Cereal Science and Technology". 3rdEdition. American Association of Cereal Chemists, 2010
R5	.Karl Kulp. "handbook of Cereal Science and Technology". 2nd Rev. Edition. CRC Press, 2000.

Online Resources

Sl.No	Website Link
1	https://www.britannica.com/science/legume
2	https://asutoshcollege.in/new-web/Study_Material/Processing_of_wheat_03042020.pdf
3	https://wingreensharvest.com/blogs/news/what-is-millets-different-type-of-millets-with-benefits-nutrition
4	https://asutoshcollege.in/new-web/Study_Material/processsing_of_rice_03042020.pdf
5	https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/parboiling